

# Izack Takazawa

Honolulu, HI | [takazawaizack@gmail.com](mailto:takazawaizack@gmail.com) | [808] 347-5733 | [linkedin.com/in/izack-takazawa-25b850330](https://www.linkedin.com/in/izack-takazawa-25b850330)

NCAA student-athlete and Data Science undergraduate with hands-on experience in high-performance computing, machine learning, and data visualization. Skilled in transforming complex datasets into actionable insights across environmental and health applications. Brings discipline, teamwork, leadership, and creativity from competitive athletics into collaborative, mission-driven technical projects. Passionate about using Data Science and AI to create a social impact within my community.

## EDUCATION

**Chaminade University of Honolulu**, Honolulu, HI— B.S. Data Science, Concentration: Health Informatics and Analytics

AUGUST 2024 - MAY 2027 EXPECTED

**St. Olaf College**, Northfield, MN — Coursework in Biology

AUGUST 2023 - MAY 2024

**Iolani School**, Honolulu, HI— High School Diploma

K-12 Graduated JUNE 2023

## TECHNICAL EXPERIENCE & PROJECTS

**NSF/LCCF Advanced Computing Student Collaborative (ACSC)**, Texas Advanced Computing Center (TACC)

NOVEMBER 2025

- Leveraging TACC supercomputers in Python to investigate how elevated PM2.5 air pollution impacts respiratory health and chronic illness rates among Hawaii residents, especially those near Kilauea volcano, leveraging sensor and API datasets from 2015 to 2025.
- Aiming to inform and protect vulnerable communities across Hawaii by highlighting air quality risks and guiding health-focused interventions.

**EPSCoR/SCIFE Undergraduate Fellow**, Chaminade University

AUGUST 2025 - PRESENT

- Delivered PM2.5 data visualizations to university faculty and stakeholders, demonstrating the ability to translate complex data into accessible insights.
- Engineering a dynamic data dashboard (Particulate Load Index) using Python and R, integrating data mining, cleaning, analysis, and visualization to quantify human exposure to airborne and ingestible particulates—including vog, plastics, and carbon emissions—across Hawai'i.

**NSF ALL-SPICE Alliance Summer Data Science Institute**, Chaminade University — Participant

MAY 2025 - JUNE 2025

- Used tech documentation, visualizations, and communications skills to present 10+ visualizations revealing that a low-glutamate, low-histamine diet drives measurable changes in inflammatory gene regions, helping to raise awareness on epigenetic trends under dietary interventions.
- Applied unsupervised learning (PCA, K-means, t-SNE) utilizing Oxford Nanopore sequencing to analyze DNA methylation changes linked to inflammatory biomarkers such as (TNF- $\alpha$ , IL-6, IL-1 $\beta$ ...) in human blood.

**Independent Research — Determining Concentrations of Glutamic Acid in Proteins Cooked for Long Periods of Time**

SEPTEMBER 2023 - PRESENT

- Presented at the Hawaii State Science Fair, showcasing that prolonged cooking significantly increases glutamate concentrations in beef, chicken, and pork, with chicken showing an interesting exponential increase in glutamate concentration.
- Utilized spectrophotometric analysis to evaluate how cooking time influences the concentration of glutamic acid alongside TNF- $\alpha$  and IL-6.

## SKILLS

**Technical Skills:** Data Exploration, Data Mining, Data Cleaning, Data Analysis, Data Visualization, High Performance Computing, Machine Learning, Statistical Modeling, GIS, Dashboard Development, Website Development, Working with Structured & Unstructured data

**Soft Skills:** Excellent Verbal and Written Communication, Critical Thinking, Analytical Problem Solving, Ability to Perform Under Pressure, Leadership, Collaboration, Adaptability, Time Management

## PROGRAMMING & TOOLS

**Programming Languages:** Python, R, SQL

**Development Tools:** TACC Super Computers, Jupyter Lab, Jupyter Notebook, RStudio, Terminal, Tableau, Shiny, MySQL, Thonny, Github, Github Desktop; Proficient in Microsoft Office and Google Workspace

## CERTIFICATIONS & HONORS

- NSF Alliance Certificate in UN Sustainable Development Goals Analytics in R program (2025)
- Certificate of Excellence, SPICE Game Health Award (2025)
- Undergraduate Research Fellowship (2025–Present)
- Dean's List (2024–Present, 4.0 GPA)
- Interscholastic League of Honolulu Honorable Mention – Baseball (2023)

## LEADERSHIP & ATHLETICS

**NCAA Division II & III Baseball**

**Student-Athlete — Chaminade University & St. Olaf College**

AUGUST 2023 – PRESENT

- Balanced 25+ hours/week of training and travel with a 4.0 GPA, demonstrating discipline, time management, and performance under pressure.

**Youth Baseball Coach & Mentor**, Honolulu, HI  
JAN 2021 – PRESENT

